

used 12V lights feed -> reverse protection -> TVS -> bulk caps -> buck + high-side switch

use: factory rheostat output, high-impedance divider 100k/22k, RC filter, BAT54S clamp

Seed XIAO RP2040. JP1 open = vehicle 5V disconnected for safe USB programming

-> R8 -> Q2 2N7000 -> R7 -> Q1 FQP27P06 P-MOSFET. R6/D5 protect gate. R10/C5 = DNP snubber

use fuse in the vehicle harness at the feed tap, not board-mounted: it must protect the wire run to the board.

